

Master 2 Artificial Intelligence : Foundations and Advanced Methods in Machine Learning

Personal website : <https://bastienroure.github.io>

Professional Experience

- Jan. - Aug. 2026 **Commissariat à l'Énergie Atomique (CEA), Research internship**, Grenoble
Development of anomaly detection methods for backdoor attacks in federated learning frameworks. Investigation of Low-Rank Adaptation (LoRA) injection mechanisms across attention weight matrices in large language models.
Keywords : Python, PyTorch, Federated Learning, LLM's, PEFT, LoRA, CUDA, Slurm.
- Jan. - July 2025 **INRIA, Research internship in a computer science laboratory**, Montbonnot
Development of unsupervised machine learning methods for learning priors, based on biomechanically accurate human models and aimed at clinically relevant medical applications.
Keywords : Python, PyTorch, Gaussian Mixture Model, Scikit-Learn, CUDA.
- May - June 2024 **Twinsight Medical, R&D Internship**, Grenoble
Design of algorithms for 3D modeling of scars following orthopedic surgery.
Keywords : C/C++, OpenGL, 3D Computer Graphics, Parallel computing.
- April 2024 **STMicroelectronics, Data Science Internship**, Grenoble
Unsupervised Machine Learning methods for microelectronic components clustering.
Keywords : Python, Machine Learning, Clustering, Data analysis.
- 2020–2023 **Tutoring, Independent**, Mathematics & Informatics — Middle school to University level.
- July 2019 **Machine Operator, AKWEL**, Champfromier
Seasonal work on a production line.

Education

- 2025 **Master 2 Artificial Intelligence : Foundation and advanced Methods in Machine Learning**
UGA - INP ENSIMAG, Grenoble
- 2024–2025 **Master 1 Master of Science in Informatics at Grenoble**
UGA - INP ENSIMAG, Grenoble
- 2021–2024 **Bachelor Mathematics & Informatics**
Grenoble Alpes University, Grenoble
- 2021 **French general Baccalauréat**, Options : Mathematics, Physics, Economy,
Saint-Julien-en-Genevois

Skills

- Informatics Machine Learning, Deep Learning, Computer Vision, Algorithmic, Programming (C, C++, Python, Bash, R, SQL, OCaml, Assembly), Data Base, Operating System, Embedded System.
- Mathematics Graph theory, Optimization, Differential modeling, Differential geometry, Statistics, Probability, Topology, Algebra, Analysis.
- Languages French (native), English (B2 - Upper intermediate), Spanish (B1 - intermediate).

Profile

- Fighting sports Boxing, Kick-boxing/K1, Krav-Maga/MMA — practicing for 6 years.
- Archery 8 years of practice — 3× Rhône-Alpes regional champion ; 5× Haute-Savoie departmental champion ; National Top 100 (Benjamin category).